

## VIKING AGCM, AB, Canada

WMO: 718880

Lat: 53.182N

Lon: 111.732W

Elev: 711

StdP: 93.07

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p) |
| 2             | -30.5      | -28.0 | -33.9                         | 0.2 | -30.4 | -31.2 | 0.2 | -28.0 | N/A                   | N/A  | N/A | N/A  | N/A                   | N/A  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.2                   | 27.6            | 16.8 | 26.1 | 16.3 | 24.5 | 15.7 | 19.2                | 25.0 | 18.0 | 23.8 | 16.9 | 22.3 | N/A                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.0                            | 13.2 | 22.6 | 15.7 | 12.2 | 20.8 | 14.7 | 11.4 | 19.6 | 57.6          | 24.8 | 53.6 | 23.7 | 50.2 | 22.5 | 24.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) N/A           | N/A  | N/A | -34.1                      | 30.8 | 1.9                | 1.6 | -35.5                                              | 32.0 | -36.6      | 33.0 | -37.7      | 33.9 | -39.1      | 35.0 |    |    |
| (5)               |      |     | -34.3                      | 21.8 | 1.8                | 1.4 | -35.6                                              | 22.7 | -36.7      | 23.5 | -37.7      | 24.3 | -39.0      | 25.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 2.6    | -11.0 | -12.1 | -5.2  | 2.9   | 10.8  | 14.6  | 16.6  | 16.1  | 11.0  | 3.9   | -5.1  | -11.9 |  |  |
|                                                                        | DBStd       | 12.27  | 8.23  | 7.86  | 7.41  | 5.28  | 4.55  | 3.15  | 2.60  | 3.24  | 4.59  | 5.02  | 7.12  | 7.56  |  |  |
|                                                                        | HDD10.0     | 3382   | 651   | 620   | 472   | 218   | 45    | 5     | 0     | 1     | 41    | 197   | 453   | 679   |  |  |
|                                                                        | HDD18.3     | 5762   | 909   | 853   | 731   | 462   | 234   | 117   | 66    | 82    | 221   | 447   | 703   | 938   |  |  |
|                                                                        | CDD10.0     | 697    | 0     | 0     | 0     | 6     | 71    | 144   | 206   | 191   | 72    | 8     | 0     | 0     |  |  |
|                                                                        | CDD18.3     | 36     | 0     | 0     | 0     | 0     | 2     | 6     | 13    | 13    | 2     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 590    | 0     | 0     | 0     | 2     | 68    | 74    | 164   | 208   | 72    | 2     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 96     | 0     | 0     | 0     | 0     | 10    | 8     | 27    | 38    | 12    | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | PrecAvg     | 409    | 22    | 12    | 16    | 30    | 41    | 70    | 74    | 58    | 34    | 17    | 19    | 18    |  |  |
| (15) Precipitation                                                     | PrecMax     | 586    | 70    | 23    | 32    | 94    | 65    | 124   | 149   | 125   | 80    | 33    | 83    | 33    |  |  |
|                                                                        | PrecMin     | 217    | 5     | 1     | 4     | 4     | 5     | 15    | 28    | 11    | 3     | 1     | 2     | 4     |  |  |
|                                                                        | PrecStd     | 85     | 16    | 7     | 8     | 21    | 16    | 28    | 32    | 26    | 23    | 9     | 19    | 9     |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 5.6   | 5.3   | 11.9  | 21.9  | 28.1  | 27.6  | 29.6  | 29.9  | 28.5  | 22.0  | 12.9  | 5.1   |  |  |
|                                                                        |             | MCWB   | 2.3   | 2.4   | 6.1   | 9.7   | 13.1  | 16.0  | 19.8  | 17.4  | 15.7  | 12.3  | 6.9   | 2.0   |  |  |
|                                                                        | 2%          | DB     | 4.0   | 2.9   | 8.0   | 18.0  | 25.4  | 25.5  | 27.0  | 27.3  | 25.4  | 17.0  | 8.2   | 2.9   |  |  |
|                                                                        |             | MCWB   | 1.4   | 0.8   | 4.2   | 8.5   | 12.3  | 15.3  | 18.9  | 17.2  | 14.1  | 9.6   | 4.3   | 0.6   |  |  |
|                                                                        | 5%          | DB     | 2.5   | 1.2   | 5.4   | 14.4  | 22.6  | 23.4  | 25.1  | 25.8  | 22.9  | 14.1  | 5.5   | 0.8   |  |  |
|                                                                        |             | MCWB   | 0.6   | -0.4  | 2.6   | 6.7   | 11.5  | 14.5  | 17.7  | 17.0  | 13.5  | 8.1   | 2.7   | -1.2  |  |  |
|                                                                        | 10%         | DB     | 0.5   | -0.5  | 3.4   | 11.5  | 20.1  | 21.6  | 23.3  | 23.7  | 19.7  | 11.6  | 3.4   | -1.6  |  |  |
|                                                                        |             | MCWB   | -1.1  | -1.8  | 1.1   | 5.3   | 10.1  | 13.7  | 16.8  | 16.4  | 12.2  | 6.7   | 1.2   | -3.0  |  |  |
|                                                                        | 0.4%        | WB     | 2.7   | 2.4   | 6.8   | 10.7  | 14.2  | 18.5  | 21.8  | 20.8  | 16.9  | 12.8  | 7.3   | 2.3   |  |  |
|                                                                        |             | MCDB   | 5.1   | 4.5   | 11.0  | 21.2  | 24.7  | 24.3  | 27.1  | 25.7  | 24.7  | 20.4  | 11.8  | 4.7   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 1.7   | 1.0   | 4.3   | 8.6   | 13.0  | 16.9  | 19.7  | 19.0  | 15.1  | 10.6  | 4.6   | 0.7   |  |  |
|                                                                        |             | MCDB   | 3.7   | 2.6   | 7.4   | 16.9  | 22.2  | 23.1  | 25.6  | 25.1  | 23.4  | 15.4  | 7.9   | 2.7   |  |  |
|                                                                        | 5%          | WB     | 0.5   | -0.3  | 2.6   | 7.1   | 12.1  | 15.7  | 18.4  | 17.7  | 13.7  | 8.7   | 2.8   | -1.2  |  |  |
|                                                                        |             | MCDB   | 2.4   | 1.1   | 5.3   | 13.8  | 20.9  | 21.2  | 24.0  | 23.7  | 21.2  | 13.3  | 5.1   | 0.7   |  |  |
|                                                                        | 10%         | WB     | -1.0  | -1.9  | 1.2   | 5.7   | 11.0  | 14.6  | 17.3  | 16.7  | 12.6  | 7.1   | 1.4   | -3.0  |  |  |
|                                                                        |             | MCDB   | 0.5   | -0.5  | 3.2   | 10.8  | 18.8  | 19.8  | 22.3  | 22.5  | 19.3  | 11.3  | 3.3   | -1.6  |  |  |
|                                                                        | 5% DB       | MDBR   | 9.1   | 9.3   | 9.0   | 10.4  | 12.6  | 11.1  | 11.2  | 12.3  | 12.3  | 10.4  | 8.4   | 8.0   |  |  |
|                                                                        |             | MCDBR  | 9.5   | 10.4  | 10.5  | 15.9  | 16.3  | 14.1  | 14.3  | 15.7  | 17.6  | 14.8  | 10.9  | 10.4  |  |  |
|                                                                        |             | MCWBR  | 7.8   | 8.7   | 7.4   | 8.2   | 6.6   | 6.1   | 7.4   | 7.2   | 8.0   | 8.2   | 7.4   | 8.6   |  |  |
|                                                                        |             | MCWBR  | 9.6   | 10.0  | 10.1  | 14.7  | 14.4  | 11.7  | 13.2  | 13.7  | 15.8  | 13.7  | 10.0  | 10.3  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR  | 8.0   | 8.5   | 7.1   | 8.0   | 6.2   | 6.0   | 7.4   | 7.0   | 7.6   | 7.9   | 7.1   | 8.8   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.253 | 0.279 | 0.301 | 0.328 | 0.347 | 0.364 | 0.360 | 0.372 | 0.311 | 0.276 | 0.251 | 0.229 |  |  |
|                                                                        | taud        |        | 2.308 | 2.288 | 2.346 | 2.369 | 2.363 | 2.352 | 2.383 | 2.327 | 2.473 | 2.543 | 2.449 | 2.350 |  |  |
|                                                                        | Ebn at Noon |        | 746   | 832   | 890   | 906   | 901   | 886   | 883   | 848   | 873   | 844   | 760   | 721   |  |  |
|                                                                        | Edh at Noon |        | 58    | 81    | 96    | 107   | 114   | 117   | 111   | 111   | 84    | 63    | 50    | 45    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.02  | 2.07  | 3.57  | 4.70  | 5.80  | 5.98  | 6.21  | 5.07  | 3.56  | 2.07  | 1.01  | 0.76  |  |  |
|                                                                        | RadStd      |        | 0.10  | 0.15  | 0.28  | 0.43  | 0.40  | 0.42  | 0.32  | 0.38  | 0.38  | 0.28  | 0.11  | 0.07  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (80 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page